



Fashion Sustainability: A Database Approach

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Goal

- Create a robust and insightful database system for analyzing sustainability practices and environmental impacts in the fashion industry
- Integrate data on companies, products, materials, factories, and sustainability practices
- Use queries to identify eco-friendly companies, sustainable materials, and ethical manufacturing processes, promoting transparency and accountability

Overall we want to offer valuable insights to support responsible decision-making for both companies and consumers. It serves as a critical tool for addressing the challenges of fast fashion and fostering a shift toward a more sustainable future in the fashion industry.



Database Structure and Data



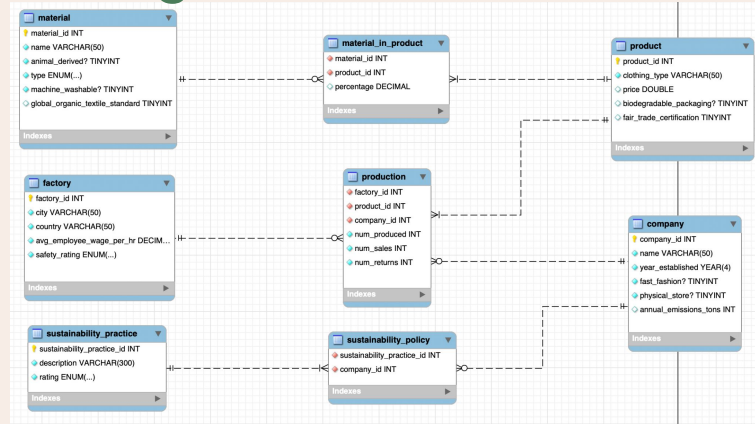
Relationships

- **Material ↔ Product:**
 - Materials are linked to products through the Material in Product table. It specifies which materials and their percentages are used in each product.
- **Product ↔ Factory ↔ Company:**
 - Tertiary relationship in the Production table.
 - Factories produce specific products along with production details like number of products produced, sold, and returned.
 - Products are tied to the companies that own them.
 - Factories are linked to companies, showing where a company's products are made.
- **Company ↔ Sustainability Practice:**
 - Companies adopt specific sustainability practices via the Sustainability Policy table.

Mock Data

Our mock data illustrates realistic scenarios in the fashion industry. It includes diverse material types, production quantities, company profiles, and sustainability initiatives. For example, EcoWear Ltd., a sustainable brand established in 2017, produces sneakers made from 100% recycled nylon. Additionally, sustainability practices such as 'Carbon Neutral Shipping' and 'Use of Recycled Materials' are adopted by companies like Sustainable Threads, emphasizing their commitment to environmental responsibility.

ER Diagram



Key Queries and Findings



Products and companies made in factories with 'High' safety and above average wages:

Clothing Type	Company	Factory City	Factory Country	Avg Wage Per Hour
Sneakers	EcoWear Ltd	Ho Chi Minh City	Vietnam	16.00
T-Shirt	Urban Chic	Barcelona	Spain	15.00
Jacket	Vintage Vibes	Barcelona	Spain	15.00
Dress	Classic Boutique	Barcelona	Spain	15.00

Which factories produce a product that uses recycled materials and is sold in biodegradable packaging?

- Factory located in Ho Chi Minh City, Vietnam

What companies use GOTS materials and make fair trade certified products?

Company	Clothing_Type	Material Name
Fashion Corp	T-Shirt	Cotton
Urban Chic	T-Shirt	Cotton

Which products are sold by which companies that implement the given sustainable practice (Carbon Neutral Shipping)?

Clothing_Type	Company
Jacket	Trendy Apparel
Sneakers	EcoWearLtd

Key Queries and Findings Cont.



What products are made out of animal-derived material?

Clothing type	material
Jacket	wool
scarf	silk

Which companies have the most 'Excellent' sustainable practices?

company	Excellent practices count
Fashion Corp	2
Trendy Apparel	2
Sustainable Threads	2
EcoWear Ltd	2

What sustainability practices does the given (Fashion Corp) company follow?

- Use of recycled materials
- Carbon offset initiatives
- Water-saving manufacturing process
- Minimal plastic packaging

Key Queries and Findings Cont.



What is the average emissions amount per product sold for each company?

company	Avg emissions per sale
Urban Chic	950
Trendy Apparel	160
EcoWear Ltd	75
Fashion Corp	48
Vintage Vibes	20
Sustainable Threads	10
ModernStyles Inc.	Unknown
Classic Boutique	Unknown

What is each company's average return rate for their products?

company	%return rate
Classic Boutique	3.33
Fashion Corp	14.84
Urban Chic	11.00
Trendy Apparel	3.40
Sustainable Threads	1.50
EcoWear Ltd	0.50
Vintage Vibes	0.00
ModernStyles Inc.	N/A

How much inventory (number of unsold or returned products) does each company have?

company	Num inventory
Fashion Corp	3115
Trendy Apparel	17
EcoWear Ltd	202
Vintage Vibes	30
ModernStyles Inc.	0
Urban Chic	111
Sustainable Threads	203
Classic Boutique	8